MPx Spin v1.1

TRIZ, FMEA, teardown and sourcing

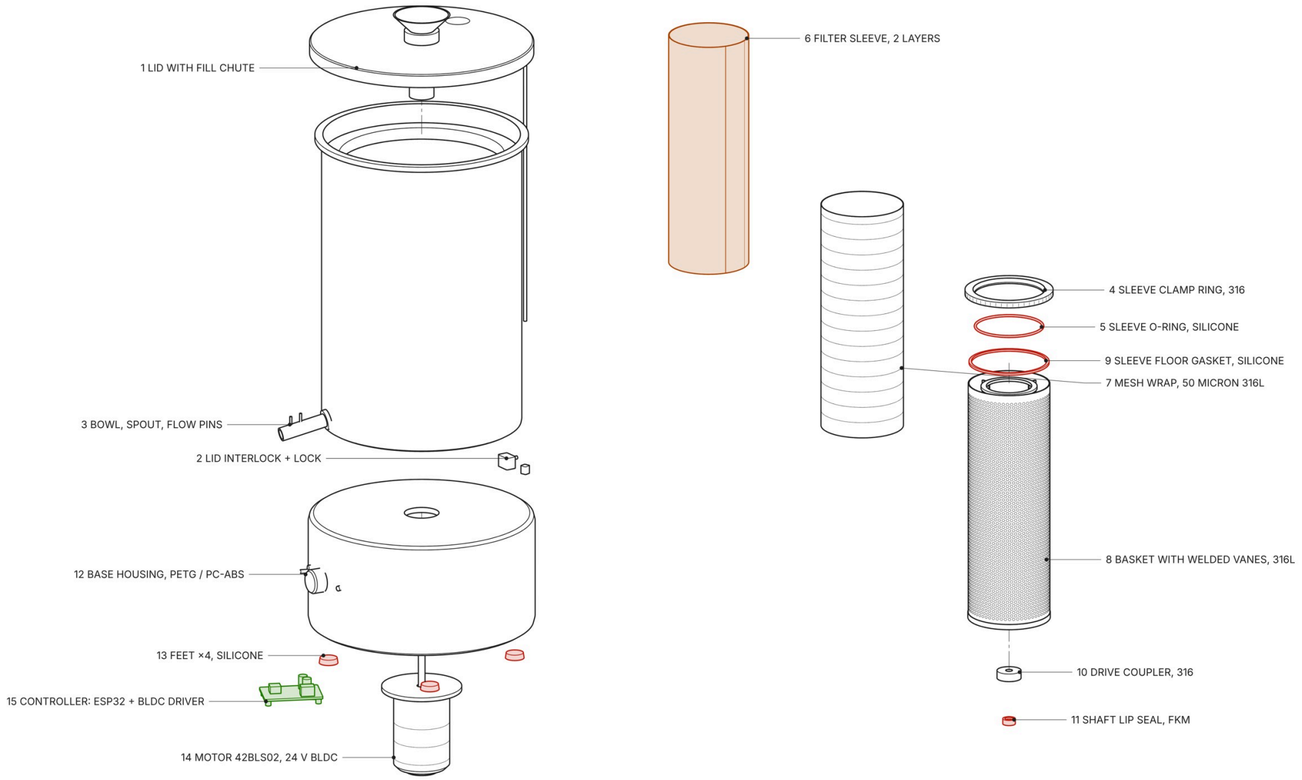
The six contradictions and four physical contradictions from Spin v1 run through the TRIZ matrix and separation principles. The ideas are tested in the calibrated model, the winners go into **v1.1**, and v1.1 is rated in a design FMEA, torn down part by part and priced from Indian suppliers.

98.9 %
caught at 1–2 µm (v1: 89 %)*

24 L/h
per 400 mL batch cycle, target 15*

228 L
per sleeve on tap water (v1: 69 L)*

10 → 0
critical failure modes, before → after



MPx Spin v1.1, exploded, item numbers as in the BOM

* simulated with the rig-calibrated model; vanes at 85 % of basket speed and the cake resistance are still assumptions

What changed from v1 to v1.1, and what it bought

Seven changes, each traced to a TRIZ principle or an FMEA action. The simulation decided between the ideas; nothing here is measured yet.

THE SEVEN CHANGES

1 Speed profile ≤ 5× face velocity
about 1,250 rpm full → 2,500 rpm draining; hall-sensor closed loop
TC2 · PC2 · principle 35, 15
FMEA 1, 6

2 Two-layer sleeve, seams 90° apart
2 × Grade 602 h class, 20 mm overlaps
TC1 · PC1 · principle 24
FMEA 1, 2

3 Vanes welded into the basket
no separate insert; sleeve slides past the vane tips
TC3 · principle 10, 5
FMEA 9

4 Silicone floor gasket
seals the sleeve's bottom edge
PC4
FMEA 2

5 Lid lock until speed = 0
solenoid pin next to the reed switch
PC4
FMEA 15

6 Flow pins in the spout
two 316 pins: batch end, drain time, end-of-life alert
principle 23 Feedback
FMEA 10, 12, 14 · req. 4.2

7 Use protocol
200 mL first flush, slow stop, 3 coupons, blank sleeve
TC1, TC4, PC3
FMEA 3, 5, 11, 13

V1 → V1.1 (SIMULATED, 25 °C, TAP WATER)

	v1	v1.1
Caught at 1–2 µm	89.3 %	98.9 %
Caught by count, 1–100 µm	94.4 %	99.4 %
Flow per 400 mL cycle	27.8 L/h	24.5 L/h
Highest face velocity	7.2×	5.0×
Sleeve life, tap	69 L	228 L
Sleeve life, borewell	17 L	57 L
Sleeve life, turbid	3.5 L	11.4 L
Flow at 5 °C	20 L/h	21 L/h
Top speed	1,500 rpm	2,500 rpm
Unbalance, 5 g off-centre	3.7 N	10 N
Parts to lift out for a sleeve	4	3
Critical FMEA modes (RN > 125)	10	0

Amber: what v1.1 gives up. A second layer halves the paper's permeance, so the cycle is slower even though the basket spins faster.

TRIZ step 1: the contradictions in matrix terms

Each contradiction from the v1 review mapped onto Altshuller's 39 parameters and looked up in the contradiction matrix (row improves, column gets worse).

	Contradiction	Improving parameter	Worsening parameter	Matrix cell: principles	Evidence from Spin v1
TC1	Capture vs flow	#30 Object-affected harmful factors	#39 Productivity	22, 35, 13, 24	602 h: 89 % at 1–2 µm; G5 slower and coarser
TC2	Speed vs capture certainty	#39 Productivity	#27 Reliability · #31 harmful side effects	1, 35, 10, 38 · 35, 22, 18, 39	1,500 rpm: 7.2× · 2,000: 12.9× · 2,500: 20×
TC3	Water speed vs simplicity	#9 Speed	#33 Ease of operation · #36 Device complexity	32, 28, 13, 12 · 10, 28, 4, 34	no vanes 1.2 L/h · vanes 28 L/h
TC4	Sample size vs sleeve life	#26 Quantity of substance	#39 Productivity	13, 29, 3, 27	count needs ≈ 20 L; borewell blinds at 17 L
TC5	Filter area vs rotor mass	#5 Area of moving object	#1 Weight of moving object	2, 17, 29, 4	more area = bigger, heavier rotor
TC6	Plastic-free wet path vs complexity	#31 Object-generated harmful factors	#36 Device complexity	19, 1, 31	no plastic means steel parts

PRINCIPLES THAT COME UP MORE THAN ONCE

35 Parameter changes
TC1, TC2 (×2)

10 Preliminary action
TC2, TC3

22 Blessing in disguise
TC1, TC2

1 Segmentation
TC2, TC6

13 The other way round
TC1, TC3, TC4

28 Mechanics substitution
TC3 (×2)

29 Pneumatics and hydraulics
TC4, TC5

4 Asymmetry
TC3, TC5

HOW THE MATRIX WAS USED

- Two parameter pairs tried where the mapping was not obvious (TC2, TC3); both cells kept.
- TC1 also checked as #23 Loss of substance ↔ #39: principles 28, 35, 10, 23. Same direction: 35 and 10 again.
- Every principle read against the spinning basket; ideas that changed the flow or capture were run in the model before being adopted (slide 6).

Matrix cells from TRIZ40 (triz40.com) contradiction matrix, checked cell by cell; parameter choices are ours

TRIZ step 2: principles into ideas (TC1–TC3)

Every suggested principle was read against the spinning basket. Green ideas are in v1.1; red ones failed a check, mostly in the simulation.

TC1 Capture vs flow

#30 Object-affected harmful factors ↔ #39 Productivity

Principles 22, 35, 13, 24

24 Intermediary

a second sleeve layer catches what the first misses and covers its seams

adopted

10 Preliminary action

flush a new sleeve with 200 mL to waste before use

adopted

22 Blessing in disguise

silt cake as an extra depth filter

not relied on

13 The other way round

make particles bigger with a coagulant

rejected: adds a chemical

TC2 Speed vs capture certainty

#39 Productivity ↔ #27 Reliability · #31 harmful side effects

Principles 1, 35, 10, 38 · 35, 22, 18, 39

35 Parameter changes

speed profile: slow while the basket is full, faster as it drains, face velocity held at 5×

adopted

10 Preliminary action

300 rpm pre-spin seats the sleeve and the water before the ramp

adopted

1 Segmentation

split the batch into small doses

rejected: pour already 10 s

18 Mechanical vibration

silicone feet isolate the residual unbalance

kept

TC3 Water speed vs simplicity

#9 Speed ↔ #33 Ease of operation · #36 Device complexity

Principles 32, 28, 13, 12 · 10, 28, 4, 34

10 Preliminary action

vanes welded into the basket; the sleeve slides past their tips

adopted

13 · 34

pleats in the sleeve act as the vanes and are thrown away with it

rejected by simulation

28 Mechanics substitution

tangential jet spins the water up

rejected: pour energy too small

32 Colour changes

dye tracer shows how far the water lags

bench test 1

TRIZ step 2 (cont.): TC4–TC6 and the physical contradictions

Where one parameter must take two opposite values, the separation principles gave the cleaner answers.

TC4 Sample size vs sleeve life

13, 29, 3, 27

- **3 Local quality:** count three Ø12 mm coupons at marked places, not the whole sleeve
- **27 Cheap short-living objects:** the sleeve is a consumable; track cost per litre
- **29 Pneumatics and hydraulics:** back-flush the sleeve to extend its life
- **13 The other way round:** coarse pre-sock in the funnel for turbid water

TC5 Filter area vs rotor mass

2, 17, 29, 4

- **17 Another dimension:** pleated sleeve: 2.3× the area in the same basket
- **2 Taking out:** keep the area, raise the head instead (speed profile)

TC6 Plastic-free wet path vs complexity

19, 1, 31

- **1 Segmentation:** bowl rolled from sheet with a welded floor, not deep-drawn
- **31 Porous materials:** perforated cage + woven mesh stay steel

PHYSICAL CONTRADICTIONS → SEPARATION PRINCIPLE → V1.1

PC1 Pores must be small (catch 1 µm) and large (pass 15 L/h)

separation in Space

two layers in series: each stays as open as 602 h, together they catch 98.9 % at 1–2 µm

PC2 The basket must spin fast (flow) and slowly (paper rating, noise)

separation in Time

speed profile: about 1,250 rpm when full, rising to 2,500 rpm as the water thins

PC3 Particles must reach the paper and float inward (PP, PE > 20 µm)

separation in Condition

slow, steady stop lets the floating layer settle onto the sleeve; count a floor coupon too

PC4 The sleeve must be sealed and removable

separation in Time + self-service (25)

spin pressure presses the 20 mm overlaps shut; floor gasket and clamp ring seal the ends; lifts out when stopped

TRIZ step 3: the ideas tested in the calibrated model

The pleated sleeve looked best on paper (2.3× the area, pleats as vanes) and lost: its walls sit closer to the axis, where the spinning water has less head.

Flow per cycle vs capture at 1–2 µm (bubble area ∝ tap-water sleeve life)



Two layers alone at fixed 1,500 rpm drop to 17 L/h and 19 L of sleeve life. It is the speed profile that pays for the second layer.

Option	L/h	Face ×	Life tap / bore L	1–2 µm
v1: 1 layer, fixed 1,500 rpm	27.8	7.2	69 / 17	89.3 %
Pleated cartridge, fixed 1,500 rpm	16.5	4.7	15 / 4	89.3 %
Pleated cartridge, 2,000 rpm	22.8	8.4	87 / 22	89.3 %
Speed profile ≤ 5×, 1 layer	36.5	5.0	281 / 70	89.3 %
Speed profile ≤ 5×, 2 layers (v1.1)	24.5	5.0	228 / 57	98.9 %
Fixed 1,500 rpm, 2 layers	17.2	7.2	19 / 5	98.9 %
Grade 5, speed profile ≤ 5×	25.4	5.0	235 / 59	78.3 %

WHY THE SPEED PROFILE WINS

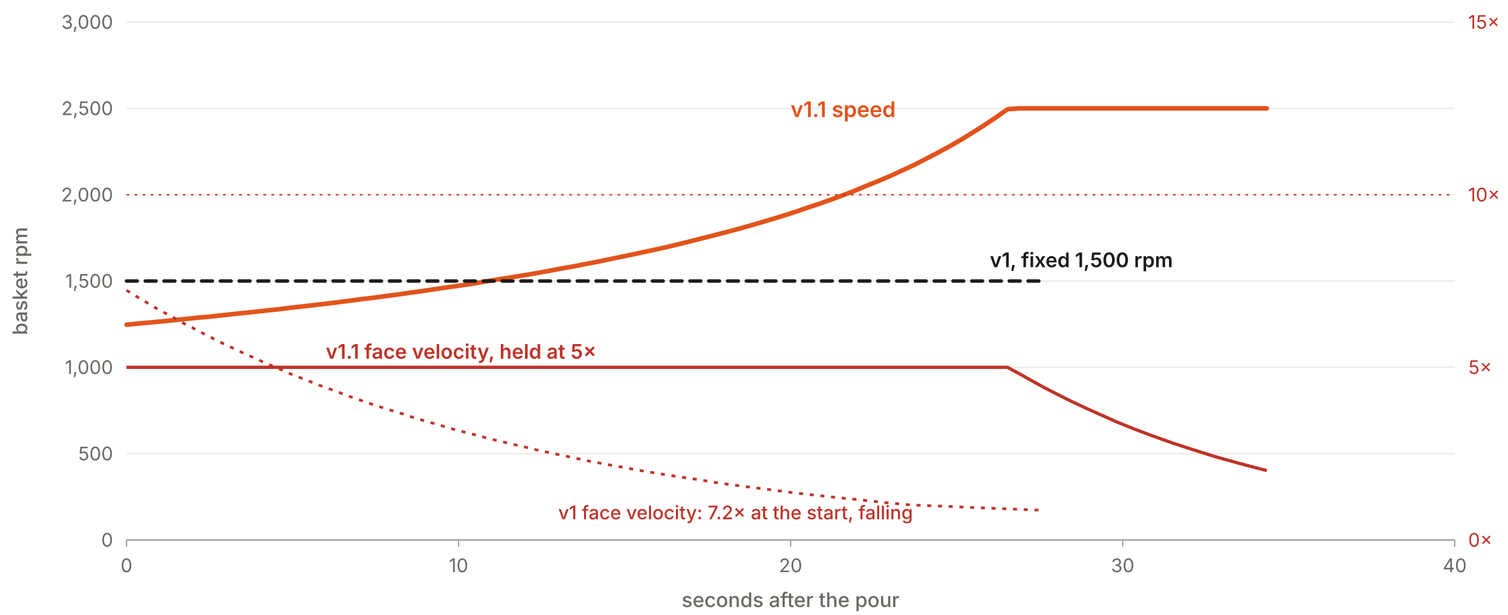
Face velocity is highest when the basket is full, so a fixed speed must be chosen for the first seconds and wastes the rest of the batch. Holding face velocity at 5× lets the basket speed up as the water thins: more flow, less paper stress, and a sleeve that keeps passing 15 L/h for much longer.

v11.mjs: pleats N = 24, depth 9 mm; two 602 h layers = Herzberg 1,500 s, penetrations multiplied; slip 0.85

The speed profile, one 400 mL batch

Separation in time: the controller picks, at every moment, the fastest speed that keeps the water's speed through the paper at 5× its rating test.

Basket speed (left) and face velocity ÷ rating test (right) during one batch



v1.1 starts at about 1,247 rpm with the basket full and reaches 2,500 rpm when about a quarter of the water is left. 90 % is through in 35 s with two layers of paper (v1: 29 s with one).

WHAT THE CONTROLLER NEEDS

- **Speed:** the motor's hall sensors, read by the ESP32 (closes the loop the rig never had).
- **Batch size:** fixed by the 400 mL funnel mark.
- **Batch end:** water between the two spout pins.
- **Profile:** a table of rpm vs time from this model; corrected by the measured drain time.

COST OF GOING FASTER

- Water at the wall reaches **151 g** near the end of a batch.
- 5 g off-centre → **10 N** at 2,500 rpm (3.7 N at 1,500).
- Basket hoop stress still only 0.74 MPa: safety factor 229×.

v11.mjs; face velocity relative to the Whatman Herzberg test (100 mL through 10 cm² under a 10 cm head)

Design FMEA: scope and scales

Pahl & Beitz §10.4. v1 rated as it stood after the first review, then re-rated with the v1.1 changes and protocol. Two failure modes are new because of v1.1.

1 Risk analysis

Failures, effects, causes; current controls

2 Risk assessment

Occurrence O, severity S, detection D (easy to detect = low D)

3 Risk number

RN = O × S × D, 1–1000. **RN > 125 is critical**

4 Risk minimisation

A design or protocol change for every critical item, then re-rate

Occurrence O

1–2	A design feature prevents it
3–4	Occasional
5–6	Likely over a year of use
7–8	Most users, often
9–10	The simulation shows it every time

Severity S

1–2	Barely noticed
3–4	Slower or less convenient
5–6	Function degraded
7–8	Wrong result, hygiene or injury risk
9–10	Untreated water passed as treated

Detection D

1–2	Obvious at once
3–4	User notices within a use
5–6	Only by inspection or a test
7–8	Hidden from the user
9–10	Cannot be detected

Scope: lid and interlock, bowl and spout, rotor (basket, vanes, mesh, sleeve, seals), drive and controller, and four use steps: pour, spin, change the sleeve, read the sleeve. **20 failure modes; 10 critical before, 0 after.** Ratings use the model's numbers where it has them (capture, flow, sleeve life, stress); the rest is our judgement for the team to challenge in the session.

Design FMEA (1/2): the ten highest risks

Sorted by RN before. Red RN is critical (> 125). New RN is the re-rating with the v1.1 change or protocol in place.

#	Part / function	Failure mode	Effect	Cause	Control (v1)	O	S	D	RN	Action in v1.1	O·S·D	New RN	Req.
1	**Sleeve paper**	Capture falls at high face velocity	1–2 µm particles pass	Rated at 1/7 of the working velocity	None	6	9	9	486	Speed profile ≤ 5× face velocity; second layer; bead test 5	2·9·6	108	6.1
2	**Sleeve seam and ends**	Water bypasses the paper	Unfiltered particles reach the jug	Seam or ends unsealed (rig K 3.2× datasheet)	Clamp ring + O-ring	6	9	8	432	20 mm overlaps pressed by spin; seams 90° apart in 2 layers; floor gasket; dye test	2·9·5	90	6.1
3	**Detection**	Cellulose fibres or stain read as plastic	False positives	Nile red uptake by cellulose	None	8	7	7	392	Blank sleeve per batch of sleeves; fibre-shape exclusion; reference counts	4·7·4	112	6.2
4	**Spent sleeve**	Binned loose or rinsed	Microplastics back to the environment	No disposal route	None	7	6	8	336	Sealed sleeve pouch in the box; return-to-lab route for the study	3·6·6	108	10.1
5	**Light polymers**	PP / PE miss the paper or land unevenly	Sample under-counts them	Inward drift beats through-flow above ≈ 20 µm	None	6	6	8	288	Slow stop; floor coupon counted; PP bead test	4·6·5	120	1.2
6	**Speed control**	Speed drifts out of the band	Low flow or uncertain capture	Open-loop PWM, no feedback	None	6	6	8	288	Hall-sensor closed loop on the ESP32; speed logged	2·6·2	24	2.1
7	**Lip seal**	Leaks into the dry base	Water on the motor and board	Wear at speed, dry running	Lip seal	4	7	7	196	Boss 14 mm above the floor; drip path out of the base; board on stand-offs	3·5·4	60	3.5
8	**Wet sleeve**	Tears or creases at 55–150 g	Bypass; fibres in the water	Wet strength over Ø2 holes	Mesh support	4	8	6	192	Two layers; inspect on lift-out; 20-batch test 7	3·8·4	96	6.1
9	**Sleeve change**	Sample touched or dropped	Contaminated sample, spilled catch	Four-step removal, wet hands	None	5	6	6	180	Three steps (vanes welded in); tweezer tabs; gloves; pouch	3·6·4	72	10.2
10	**Speed profile** new in v1.1	Starts fast with a full basket	Face velocity ≈ 20× at the start	Firmware fault or wrong batch size	— (new in v1.1)	3	7	7	147	Start capped at 1,300 rpm until the flow pins see water; batch size fixed by the funnel	2·7·4	56	6.1

Design FMEA (2/2): the remaining modes

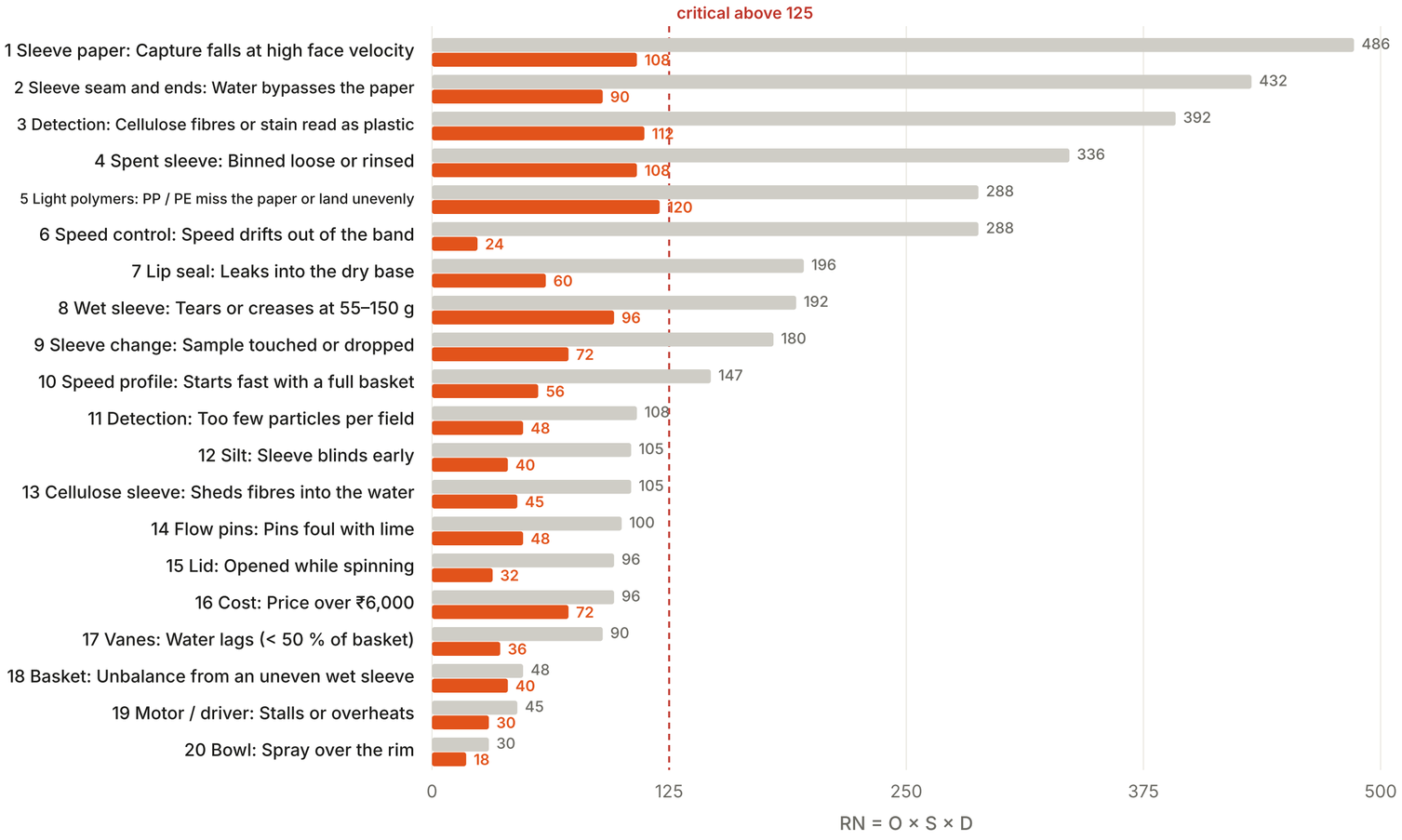
Lower risks, kept so the actions and the bench tests cover them too.

#	Part / function	Failure mode	Effect	Cause	Control (v1)	O	S	D	RN	Action in v1.1	O·S·D	New RN	Req.
11	**Detection**	Too few particles per field	Count meaningless	Sparse sample on a large sleeve	None	9	6	2	108	Coupons after ≥ 20 L on one sleeve	4·6·2	48	6.2
12	**Silt**	Sleeve blinds early	Flow under 15 L/h	Borewell / turbid water	None	7	5	3	105	Life 228 / 57 / 11 L; alert from drain time	4·5·2	40	9.1
13	**Cellulose sleeve**	Sheds fibres into the water	Particles in the output	Paper rubbing on the mesh	None	5	3	7	105	200 mL first flush to waste	3·3·5	45	3.2
14	**Flow pins** new in v1.1	Pins foul with lime	Wrong batch-end signal	Scale on 316 pins	— (new in v1.1)	5	4	5	100	Pins wiped at each sleeve change; timed fallback profile	3·4·4	48	4.2
15	**Lid**	Opened while spinning	Hand near a basket at 7.9 m/s	Interlock bypassed	Reed switch	3	8	4	96	Solenoid lock until hall speed = 0	2·8·2	32	safety
16	**Cost**	Price over ₹6,000	Not affordable	Steel parts, BLDC drive	None	8	6	2	96	Rolled parts, printed base; production costing next	6·6·2	72	11.1
17	**Vanes**	Water lags (< 50 % of basket)	Flow under 15 L/h	Too few or short vanes	4-vane insert	5	6	3	90	Welded vanes; bench test 1; profile adapts to drain time	3·6·2	36	2.1
18	**Basket**	Unbalance from an uneven wet sleeve	Vibration, noise	Uneven cake or sleeve	None	6	4	2	48	Pre-spin seats the sleeve; feet; 5 g → 10 N at 2,500 rpm checked	4·5·2	40	8.1
19	**Motor / driver**	Stalls or overheats	No flow	Start torque, no current limit	None	3	5	3	45	Soft start; driver current limit	2·5·3	30	2.1
20	**Bowl**	Spray over the rim	Wet bench	Water leaves the cage at ≈ 7 m/s	Skirt	5	3	2	30	Skirt kept; lid closes the rim	3·3·2	18	—

FMEA result: 10 critical modes before, 0 after

Most of the drop comes from detection (D): the hall-sensor loop, the flow pins, the dye test and the blank sleeve turn hidden failures into visible ones.

Risk number per failure mode, before (grey) and after (orange)



TOTALS

3,760 → 1,255

sum of RN over 20 modes (67 % lower)

10 → 0

critical modes (RN > 125)

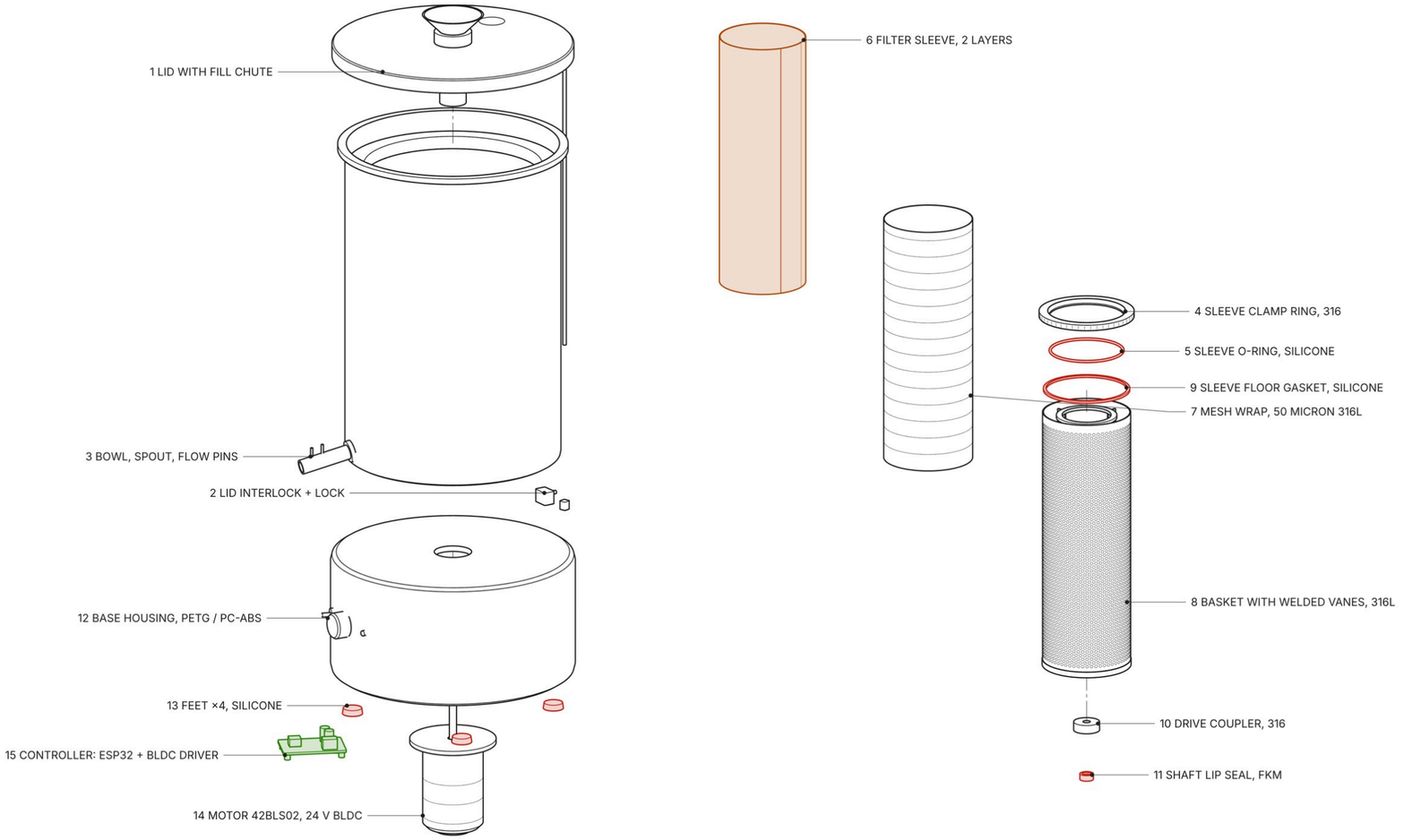
STILL THE BIGGEST AFTER THE ACTIONS

- **Light polymers, pp / pe miss the paper or land unevenly** (RN 120): Slow stop.
- **Detection, cellulose fibres or stain read as plastic** (RN 112): Blank sleeve per batch of sleeves.
- **Sleeve paper, capture falls at high face velocity** (RN 108): Speed profile ≤ 5× face velocity.
- **Spent sleeve, binned loose or rinsed** (RN 108): Sealed sleeve pouch in the box.

Severity is rarely lowered by these actions; it takes a different concept to make a missed particle less serious

Teardown: how v1.1 goes together

Fifteen parts. Item numbers match the BOM. The vane insert is gone (welded into item 8) and the floor gasket (9) and lid lock (2) are new.



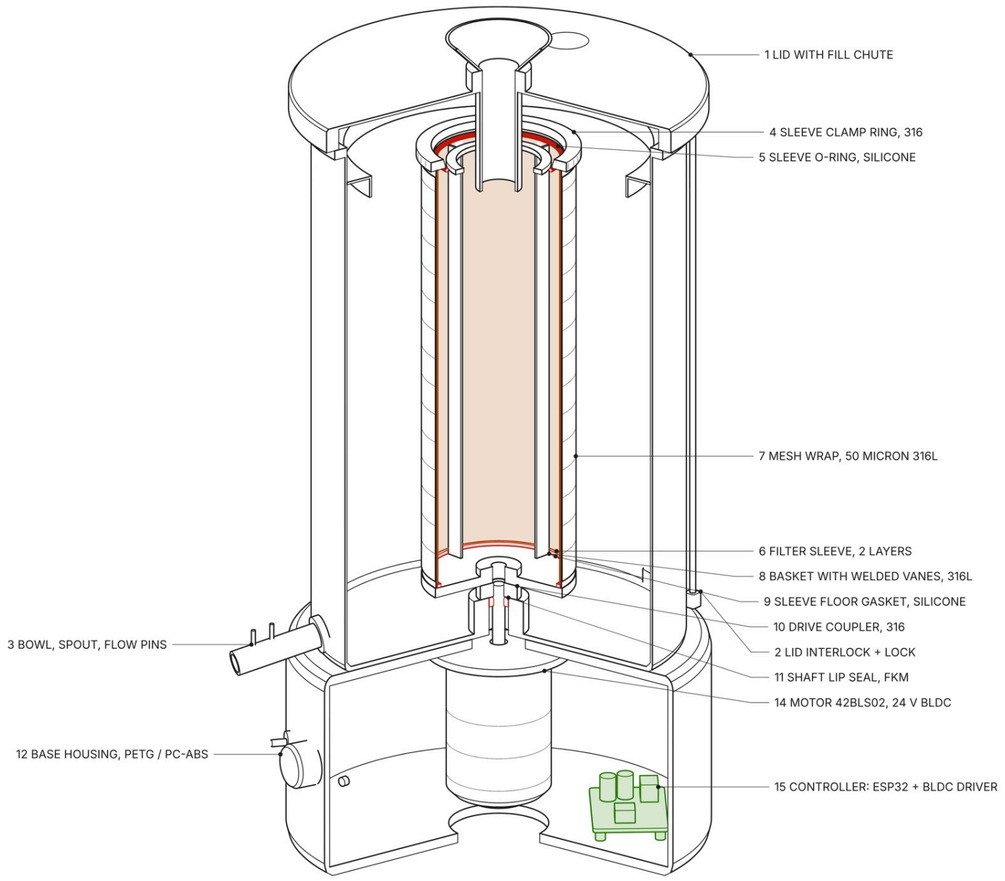
PARTS, MATERIALS, MASS AS DRAWN

1	**Lid with fill chute** 304, 1.2 mm; glass window · spin/press, TIG	475 g
2	**Lid interlock + lock** reed switch, 24 V solenoid · buy	25 g
3	**Bowl, spout, flow pins** 304, 1.0 mm drawn (0.6–0.8 built) · roll, seam + TIG	1,311 g
4	**Sleeve clamp ring** 316 · CNC turn	39 g
5	**Sleeve O-ring** silicone 60 ShA · buy	0.4 g
6	**Filter sleeve, 2 layers** cellulose, 602 h class · cut from circles	6.0 g
7	**Mesh wrap, 50 µm** 316L 300-mesh twill · cut, spot-weld	9.1 g
8	**Basket with welded vanes** 316L perforated 0.6 mm · laser, roll, TIG	245 g
9	**Sleeve floor gasket** silicone sheet 1.5 mm · knife cut	0.6 g
10	**Drive coupler** 316 · CNC turn	11 g
11	**Shaft lip seal** FKM 5 × 12 × 4 · buy	0.6 g
12	**Base housing** PETG / PC-ABS · FDM print / mould	206 g
13	**Feet ×4** silicone · buy	3.0 g
14	**Motor 42BLS02** 24 V BLDC, hall sensors · buy	430 g
15	**Controller** ESP32 + ZS-X11H · buy, perfboard	75 g
	Total as drawn	**2.84 kg**

Teardown drawing from dims.mjs + v1.1 changes; masses from the drawn geometry (Pappus volumes × density). Illustrative, not for manufacture

Teardown: cut open, and where the mass and the risk sit

The drawn bowl and lid are thick for drawing clarity: built from 0.6–0.8 mm sheet the unit loses about 0.6 kg. The spinning parts weigh about 0.3 kg.



MASS BY GROUP (AS DRAWN)



Housing and bowl	1,994 g · items 1, 3, 12, 13
Rotor (spinning)	312 g · items 4, 5, 6, 7, 8, 9, 10
Drive and electronics	531 g · items 2, 11, 14, 15

INTERFACES THAT CARRY THE FMEA'S TOP RISKS

Interface	What can go wrong	
Sleeve top ↔ clamp ring + O-ring (4, 5, 6)	bypass at the top edge	FMEA 2
Sleeve bottom ↔ floor gasket (6, 9, 8)	bypass at the floor	FMEA 2
Sleeve overlap ↔ mesh ↔ cage (6, 7, 8)	seam bypass, tears over Ø2 holes	FMEA 2, 8
Shaft ↔ lip seal ↔ boss (11, 3, 14)	leak into the dry base	FMEA 7
Lid tab ↔ reed + lock (1, 2)	opening while spinning	FMEA 15
Spout pins ↔ controller (3, 15)	wrong batch-end signal	FMEA 14

Mass groups from the Teardown tab of MPx_Spin_v11_BOM_Sourcing.xlsx

Sourcing: one prototype kit is about ₹22.3k

Bought parts are priced from Indian listings found in Oct 2026; made parts are one-off Bangalore job-shop estimates to be replaced by quotes.

#	Part	Make / buy	₹ per kit	Basis / supplier
1	Lid, chute, window	Make	₹2,500	fabricator, Peenya / Jigani
2	Interlock + lock	Buy	₹420	Robocraze reed ₹28 + solenoid
3	Bowl, spout, pins	Make	₹3,000	fabricator
4	Clamp ring	Make	₹900	CNC, Peenya
5	O-ring	Buy	₹60	O-ring kit
6	Sleeve (4 × 602 h circles)	Buy	₹543	Cytiva via distributor
7	Mesh 316L 300-mesh	Buy	₹1,000	IndiaMART ₹50–450/sq ft
8	Basket + vanes	Make	₹4,200	perforated sheet + CNC + TIG
9	Floor gasket	Buy	₹350	silicone sheet
10	Coupler	Make	₹700	CNC, 316 bar ≈ ₹240/kg
11	Lip seal FKM	Buy	₹250	seal stockist
12	Base (PETG print)	Make	₹1,500	print service
13	Feet	Buy	₹120	hardware
14	Motor 42BLS02	Buy	₹2,080	robu.in
15	ESP32 + ZS-X11H + panel	Buy	₹2,104	Techiesms ₹405, Probots ₹1,299
16	24 V 5 A supply	Buy	₹807	Robocraze ₹707 + cable
17	Fasteners	Buy	₹300	local
18	Electropolish + passivate	Buy	₹1,500	finishing shop

Green: price from a live listing. Black: estimate (yellow cells in the workbook).

BUDGET

One kit	**₹22,334**
Two kits	**₹44,668**
Sleeve stock, 100 circles = 25 sleeves	**₹13,586**
Total	**₹58,254**
PD-298 cap	**₹50,000**

Over the cap by about ₹8k. One full kit plus a spare rotor (items 4–10, ≈ ₹7.8k) and the sleeve stock comes to about ₹43.7k, under the cap.

BIGGEST COSTS

- **Welded steel parts** (items 1, 3, 8): ₹9.7k, 43 % of a kit. Quotes from two fabricators first.
- **Drive and electronics** (14–16): ₹5k, all from live listings.
- **Sleeves:** ₹543 each from imported 602 h circles.

Sourcing: the sleeve is the part to solve

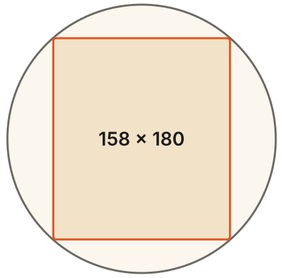
602 h is only listed as circles up to 240 mm, so one two-layer sleeve uses four circles and costs about ₹543. That makes paper, not electricity, the running cost.

SLEEVE COST PER LITRE (SIMULATED SLEEVE LIFE)

Water	Sleeve life	₹ per litre	Sleeves / year at 3 L/day	₹ / year
Tap	228 L	₹2.39	5	₹2,613
Borewell	57 L	₹9.54	19	₹10,451
Turbid surface	11 L	₹47.72	96	₹52,256

For comparison, the electricity for a batch is about 0.01 Wh. Life comes from the speed profile; cost per sleeve has to come from buying paper as sheets or rolls.

HOW A 240 MM CIRCLE BECOMES A SLEEVE



- Each circle gives one 158 × 180 mm strip.
- Two strips make one layer (188.5 mm round + two 20 mm overlaps).
- Two layers, seams turned 90°: four circles per sleeve.

RFQS TO SEND THIS WEEK

1. **Cytiva India (Whatman)**
   602 h in sheets or rolls, ≥ 210 × 180 mm; price per m² at 100 sleeves; wet-strength data
2. **Indian lab-paper makers**
   a < 2 µm cellulose paper in rolls with Herzberg time and retention data; test against 602 h
3. **Perforated-sheet makers (Mumbai)**
   0.6 mm 316L, Ø2 holes, 3.1 mm 60° pitch, 300 × 300 cut piece, mill certificate
4. **Wire-cloth suppliers**
   300-mesh 316L twill, 1 sq ft, aperture certificate
5. **Two Bangalore fabricators**
   items 1, 3, 8 from the teardown drawing; roll + seam weld + TIG; electropolish
6. **Robu / Probots**
   confirm 42BLS02 hall leads and shaft Ø5; ZS-X11H speed-pulse output

Whatman 602 h list prices: scientificfilters.com; Grade 42 sheets (46 × 57 cm) exist but are coarser (2.5 µm) and slower

Requirements: what v1.1 moves

Only the requirements whose status changed or that v1.1 puts under more pressure; the other twelve stay as in the v1 review.

#	Requirement	v1		v1.1	Why
6.1	Removal ≥ 90 % by count, 1 µm–5 mm	Partial	→	Pass · sim	99.4 % by count, 98.9 % at 1–2 µm, face ≤ 5×
6.2	Detection false +/− ≤ 10 %	At risk	→	Partial	blank sleeve, coupons, fibre exclusion; still untested
2.1	Flow ≥ 15 L/h	Pass · sim	→	Pass · sim	24 L/h per cycle (v1: 28)
9.1	Filter rated ≥ 100 L or 3 months	Partial	→	Pass · sim	228 L tap; 57 L borewell
4.2	Alert near filter end-of-life	Not yet tested	→	By design · untested	drain time from the spout pins
7.1	Filter swap tool-free ≤ 30 s	At risk	→	Partial	three steps instead of four; not timed
8.2	Usable at 5–45 °C	Pass · sim	→	Pass · sim	21–26 L/h
10.1	Disposal pathway for spent filters	Not yet tested	→	By design · untested	sealed sleeve pouch; return route for the study
11.1	Retail price ₹1,500–6,000	At risk	→	At risk	prototype ₹22k per kit; production not costed
8.1	Noise ≤ 60 dB(A)	Not yet tested	→	Not yet tested	2,500 rpm top speed raises the risk; measure

All Tier 1 requirements now pass in simulation except detection (6.2), which needs the blank-sleeve and reference-count tests. Price (11.1) is the open commercial risk.

What still rests on assumptions

The calibration fixed the rig's paper and water speed. v1.1's gains depend on four numbers no one has measured yet; each bench test moves one of them.

Vanes reach 85 % of basket speed

v1.1 still makes 15 L/h down to about 48 % (18 L/h at 55 %, 12 L/h at 40 %)

Test 1: strobe or tracer, rig with retrofit vanes

Two layers act as two stages

capture 1–2 µm 98.9 % assumes the second layer catches 89 % of what the first lets through

Test 5: spiked beads through one and two layers

Capture holds up to 5× face velocity

the rating curve is from a test at 1/5 of the working velocity

Test 5 at 0, 1,500 rpm and the profile

Silt cake $\alpha = 10^{13}$ m/kg

sleeve life 228 / 57 / 11 L scales with 1/α

Test 4: borewell water, flow vs litres

BENCH TESTS, IN ORDER

1. **Water speed, with and without vanes**
   Week 1
   settles 2.1 · FMEA 17
2. **Dye through seams, ends, floor gasket**
   Week 1
   settles 6.1 · FMEA 2
3. **Flow vs speed, 1 and 2 layers of 602 h**
   Week 2
   settles 2.1 · 6.1
4. **Borewell run: flow vs litres**
   Week 3
   settles 9.1 · FMEA 12
5. **Spiked beads at 0 / 1,500 / profile**
   Week 2–3
   settles 6.1 · FMEA 1, 5
6. **Blank sleeve under Nile red**
   Week 1
   settles 6.2 · FMEA 3
7. **20 batches on one sleeve: tears, noise**
   Week 4
   settles 3.5, 8.1 · FMEA 8, 18
8. **Time a sleeve change**
   Week 4
   settles 7.1 · FMEA 9

Next steps

Order the long-lead items now, build the rig retrofit while they arrive, and use the first two weeks of tests to decide on the paper and the vanes.

WEEK 1

Order

- 602 h circles (100) and RFQ for rolls
- Motor, driver, ESP32, supply
- Perforated sheet + 300-mesh

WEEK 1–2

Rig retrofit

- Vanes glued into the rig
- Tests 1, 2, 6 on the rig
- Decide on vanes

WEEK 2–3

Fabricate v1.1

- Quotes for items 1, 3, 8
- Print the base
- Wire ESP32 + driver; hall loop

WEEK 3

Firmware

- Speed profile table
- Flow pins, interlock, lock
- Logging over USB

WEEK 4

Test v1.1

- Tests 3, 4, 5, 7, 8
- Measured values into the model
- Re-rate the FMEA

THEN

Detail design

- Production costing (11.1)
- Noise (8.1), drop (3.5)
- Detection study with C05

DECISIONS FOR THE TEAM

- **Accept 2,500 rpm?** It is what makes two layers affordable in flow; it raises unbalance and noise (8.1).
- **Two layers by default,** or one layer for tap water and two for sampling runs?
- **Build one kit + a spare rotor** (within the ₹50k cap) or ask for more budget for two full kits?
- **Sleeve paper:** wait for 602 h rolls, or start with cut circles and accept four circles per sleeve?